STAT6611

- Lecture 08 (R, Sept 17)

- Homework 1 assigned due F, Sept 18 (F this week)
  by midnight in gradescope/canvas
  (practice on g.i. 2 matrix results)

- Homework 2 assigned due F, Oct 2 by midnight (2 weeks)
  (practice on LSE (without constraints))

- My office hours: TW 12:30-1:30 PM
  & R 1-2 PM

  TA Ben: R 9:30-10:30 AM
  in Snedecor 2207

2.9 Constrained Linear Models

In Models I—VII, $\underline{\beta}$ can assume any value in the Euclidean p-space. We now consider the case where $\underline{\beta}$ is restricted to the subspace of R^p consisting of $\underline{\beta}$ values that satisfy

$$A\underline{\beta} = \underline{b},$$

where A is a given $q \times p$ matrix of rank $q^* \geq 1$ and $\underline{b}$ is a given $q \times 1$ vector. To avoid trivialities, we assume that the linear system $A\underline{\beta} = \underline{b}$ is consistent. We will refer to those modified versions of Models I—VII in which $\underline{\beta}$ is restricted to this subspace as Models Ia—VIIa.

Definition. Under Models IIa—VIIa, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Lemma 2.9. *Under Models IIa—VIIa, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if there exists a vector $\underline{k}$ such that*

$$c = \underline{k}'\underline{b} \quad \text{and} \quad \underline{\lambda}' = \underline{a}'X + \underline{k}'A.$$

Proof. $E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$ ← constraint

$E(\underline{y}) = X\underline{\beta}$ → $\iff c + \underline{a}'X\underline{\beta} = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$

re-write → $\iff c + (\underline{a}'X - \underline{\lambda}')\underline{\beta} = 0$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$

$\Leftrightarrow$ since $\underline{\beta} \in \mathbb{R}^p$ satisfying $A\underline{\beta} = \underline{b}$ must equal $A^-\underline{b} + [I - A^-A]\underline{z}$ for some $\underline{z}$, need

$c + (\underline{a}'X - \underline{\lambda}')(A^-\underline{b} + [I - A^-A]\underline{z}) = 0$ for all $\underline{z}$

$\Leftrightarrow c + (\underline{a}'X - \underline{\lambda}')A^-\underline{b} + (\underline{a}'X - \underline{\lambda}')[I - A^-A]\underline{z} = 0$ for any $\underline{z}$

$\Leftarrow c = \tilde{\underline{k}}'\underline{b}$ for $\tilde{\underline{k}}' = (\underline{\lambda}' - \underline{a}'X)A^-$ } ①

$(\underline{\lambda}' - \underline{a}'X) = (\underline{\lambda}' - \underline{a}'X)A^-A = \tilde{\underline{k}}'A$

$\Leftrightarrow c = \underline{k}'\underline{b}$ & $\underline{\lambda}' = \underline{a}'X + \underline{k}'A$ for some $\underline{k}$ } ②

Note ① ⇒ ②

Need to show ② ⇒ ①

It remains to show that if $c = \underline{k}'\underline{b}$ and $\underline{\lambda}' = \underline{a}'X + \underline{k}'A$ for some $\underline{k}$, then ②

$c = (\underline{\lambda}' - \underline{a}'X)A^-\underline{b}$ and $\underline{\lambda}' = \underline{a}'X + (\underline{\lambda}' - \underline{a}'X)A^-A$. ← ①

First, $(\underline{\lambda}' - \underline{a}'X)A^-\underline{b} = \underline{k}'AA^-\underline{b} = \underline{k}'\underline{b} = c$ ↑ by ②

$\underline{k}'A$ under ② ; this must be $\underline{b}$ since $A\underline{\ell} = \underline{b}$ has solution so $\underline{\ell} = A^-\underline{b}$

Next,

$(\underline{\lambda}' - \underline{a}'X) - (\underline{\lambda}' - \underline{a}'X)A^-A = \underline{k}'A - \underline{k}'AA^-A = 0$

$\underline{k}'A$ under ② ; $\underline{k}'A$; A

Definition. Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, if there exists a linear estimator $c + \underline{a}'\underline{y}$ such that $E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Theorem 2.11. *Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}$.*

proof: $\underline{\lambda}'\underline{\beta}$ is estimable $\Leftrightarrow$ $\underline{\lambda}' = \underline{a}'X + \underline{k}'A$ holds for some $\underline{a}$ & $\underline{k}$

$\Leftrightarrow \underline{\lambda}' = (\underline{a}'\ \underline{k}')\begin{pmatrix} X \\ A \end{pmatrix}$

$= \underline{t}'\begin{pmatrix} X \\ A \end{pmatrix}$ $\underline{t}' = (\underline{a}', \underline{k}')$

$\Leftrightarrow \underline{\lambda}' \in R\begin{pmatrix} X \\ A \end{pmatrix}$

Note: Recall $\underline{\lambda}'\underline{\beta}$ is estimable with No constraints means $\underline{\lambda}' \in R(X)$ (i.e. $\underline{\lambda}' = \underline{a}'X$) so any $\underline{\lambda}'\underline{\beta}$ estimable without constraints remains estimable with constraints $A\underline{\beta} = \underline{b}$ since $\underline{\lambda}' \in R(X) \subset R\begin{pmatrix} X \\ A \end{pmatrix}$

that is, a constraint $A\underline{\beta} = \underline{b}$ (parameter assumptions) ENLARGES what $\underline{\lambda}'\underline{\beta}$ are estimable because $R\begin{pmatrix} X \\ A \end{pmatrix}$ is larger $R(X)$ (more rows)

Say $X = \underset{2\times 3}{\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}}$, $\underline{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \in \mathbb{R}^3$ $\text{rank}(X) = 2 < 3$ can't estimate any $\underline{\lambda}'\underline{\beta}$

If we add constraint $A\beta = 0$ for $A = [0, 1, 1], b = 0$ (i.e. $\beta_2 = -\beta_3$), then $\begin{pmatrix} X \\ A \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ has rank 3

3×3

So any $\lambda'\beta$ with $\lambda' \in \mathbb{R}^3$ is estimable.

We now consider the least squares estimation of an estimable function $\underline{\lambda}'\underline{\beta}$ under Models IIa—VIIa.

Definition. Under Models IIa—VIIa, a least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta}) \equiv (\underline{y} - X\beta)'(\underline{y} - X\beta)$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

The Lagrangian function for the problem of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$, subject to the constraints $A\underline{\beta} = \underline{b}$, is

$$L(\underline{\beta}, \underline{\rho}) = Q(\underline{\beta}) + 2\underline{\rho}'(A\underline{\beta} - \underline{b}).$$

We have

$$\frac{\partial L}{\partial \underline{\beta}} = -2X'\underline{y} + 2X'X\underline{\beta} + 2A'\underline{\rho} \quad \text{and} \quad \frac{\partial L}{\partial \underline{\rho}} = 2(A\underline{\beta} - \underline{b}).$$

Equating these partial derivatives to zero gives

$$X'X\underline{\beta} + A'\underline{\rho} = X'\underline{y} \quad \text{and} \quad A\underline{\beta} = \underline{b}.$$

Thus, a necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, is that the equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

have a solution whose "first" component is $\widehat{\underline{\beta}}$. We refer to these equations as the constrained normal equations.

The following lemma can be used to establish the consistency of the constrained normal equations.

restatement of earlier result $BW=D$ iff $C(D) \subset C(B)$

Lemma 2.10. *The matrix equation $BW = D$ is consistent, that is, has a solution for W, if and only if $\mathcal{C}(D) \subset \mathcal{C}(B)$ if and only if $\mathcal{C}^{\perp}(B) \subset \mathcal{C}^{\perp}(D)$ that is, if and only if $\underline{a}'D = \underline{0}'$ for every vector $\underline{a}$ such that $\underline{a}'B = \underline{0}'$.*

($\Rightarrow$) If $D=BW$, then $\underline{a}'B=\underline{0}' \Rightarrow \underline{a}'D=\underline{a}'BW=\underline{0}'$

($\Leftarrow$) Note $(I-P_B)B=B-P_BB=B-B=0$

then $(I-P_B)D=0$ holds by assumption

So, $D=\underbrace{(I-P_B)D}_{0}+P_BD=B(B'B)^{-}B'D$

$=BW$, $W=(B'B)^{-}B'D$

Theorem 2.12. *The constrained normal equations are consistent.*

Proof. Take $\underline{a} = \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix}$ to be any vector such that

$$\underline{a}' \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1', \underline{a}_2') \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1'X'X + \underline{a}_2'A, \underline{a}_1'A') = \underline{0}'.$$

By Lemma 2.10, it suffices to show that

$$\underline{a}' \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix} = (\underline{a}_1', \underline{a}_2') \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix} = \underline{a}_1'X'\underline{y} + \underline{a}_2'\underline{b} = 0.$$

Theorem 2.13. *$Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, if and only if the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.*

Proof. Let $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ represent any solution to the constrained normal equations, and let $\widetilde{\underline{\beta}}$ represent any solution to $A\underline{\beta} = \underline{b}$.

Then, $X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'\underline{y}$, $A\widehat{\underline{\beta}} = \underline{b}$, and $A\widetilde{\underline{\beta}} = \underline{b}$. We have

$$Q(\widetilde{\underline{\beta}}) = (\underline{y} - X\widetilde{\underline{\beta}})'(\underline{y} - X\widetilde{\underline{\beta}}) = [(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'[(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]$$

$$= Q(\widehat{\underline{\beta}}) + [X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'[X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})] + 2[X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'(\underline{y} - X\widehat{\underline{\beta}}).$$

Furthermore, if $\underline{\beta}^*$ is any other value of $\underline{\beta}$ at which $Q(\underline{\beta})$ assumes a minimum, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, then $Q(\underline{\beta}^*) = Q(\widehat{\underline{\beta}})$. Thus, $[X(\widehat{\underline{\beta}} - \underline{\beta}^*)]'[X(\widehat{\underline{\beta}} - \underline{\beta}^*)] = 0$. This implies that $X(\widehat{\underline{\beta}} - \underline{\beta}^*) = \underline{0}$, or $X\underline{\beta}^* = X\widehat{\underline{\beta}}$. Then,

$$X'X\underline{\beta}^* + A'\underline{\rho} = X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'\underline{y},$$

so that $\begin{bmatrix} \underline{\beta}^* \\ \underline{\rho} \end{bmatrix}$ is also a solution to the constrained normal equations.